WORKED SOLUTION

STEP III 2000, Question 7

Pure • Inequalities and Limits

STEP III 2000, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q7

The key idea

Both inequalities come from comparing a product with something we already understand. Expanding $\left(1+\frac{1}{n}\right)^n$ by the binomial theorem gives terms that look just like $\frac{1}{r!}$, only a little smaller. For the product $\mathrm{P}(n)$, the AM–GM inequality turns a product of n factors into an nth power of their average. The factors are $1+\frac{1}{2^k}$, and their average is just under $1+\frac{1}{n}$, which links straight back to the first part.

Solution

The bound $\left(1+\frac{1}{n}\right)^n < \mathrm{e}$

By the binomial theorem,

$$\left(1+\frac{1}{n}\right)^n = \sum_{r=0}^{n}\binom{n}{r}\frac{1}{n^r} = \sum_{r=0}^{n}\frac{1}{r!}\cdot\frac{n(n-1)(n-2)\cdots(n-r+1)}{n^r}.$$

The second fraction is a product of r factors,

$$\frac{n(n-1)\cdots(n-r+1)}{n^r} = 1\cdot\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-1}{n}\right),$$

each lying in $(0,1]$ because $0 \le j \le r-1 \le n-1$. So the product is at most 1, and every term of the expansion is at most $\frac{1}{r!}$. Hence

$$\left(1+\frac{1}{n}\right)^n \le \sum_{r=0}^{n}\frac{1}{r!} < \sum_{r=0}^{\infty}\frac{1}{r!} = \mathrm{e},$$

where the last inequality is strict because the omitted terms $\frac{1}{(n+1)!}, \frac{1}{(n+2)!}, \ldots$ are all positive. This holds for every positive integer n.

The bound $\mathrm{P}(n) < \mathrm{e}$

Write $\mathrm{P}(n) = a_1 a_2 \cdots a_n$ with $a_k = \dfrac{2^k + 1}{2^k} = 1 + \dfrac{1}{2^k}$, which are all positive, so AM–GM applies. The arithmetic mean is

$$\frac{a_1 + \cdots + a_n}{n} = \frac{1}{n}\left(n + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n}\right) = 1 + \frac{1 - 2^{-n}}{n},$$

by summing the geometric series $\frac{1}{2} + \cdots + \frac{1}{2^n} = 1 - 2^{-n}$. Since $1 - 2^{-n} < 1$, this mean is strictly less than $1 + \frac{1}{n}$. So AM–GM gives

$$\mathrm{P}(n)^{1/n} \leq 1 + \frac{1 - 2^{-n}}{n} < 1 + \frac{1}{n}.$$

Both sides are positive, so raising to the nth power keeps the inequality:

$$\mathrm{P}(n) < \left(1 + \frac{1}{n}\right)^n < \mathrm{e},$$

using the first part for the second inequality. This holds for every positive integer n.

> **Ask yourself** Where does the strict inequality come from? AM–GM only gives $\leq$.

The strictness comes from $1 - 2^{-n} < 1$, not from AM–GM. In fact, for $n \geq 2$ the factors a_k are all different, so AM–GM is itself strict. But when $n = 1$ there is only one factor, and AM–GM is an equality ($\frac{3}{2} = \frac{3}{2}$). The step $1 + \frac{1-2^{-n}}{n} < 1 + \frac{1}{n}$ is strict in every case, so the argument covers all n.

The limit

Each factor $a_{n+1} = 1 + 2^{-(n+1)}$ is greater than 1, so $\mathrm{P}(n+1) = a_{n+1}\mathrm{P}(n) > \mathrm{P}(n)$. The sequence $\mathrm{P}(n)$ is therefore strictly increasing, and it is bounded above by e. An increasing sequence that is bounded above converges, so $\mathrm{P}(n)$ tends to a limit L.

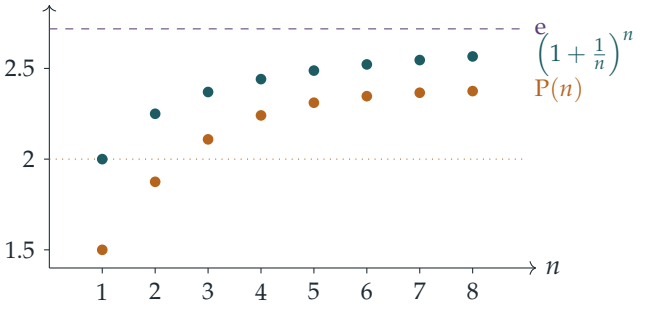
For the upper bound: every $\mathrm{P}(n) < \mathrm{e}$, so the limit satisfies $L \leq \mathrm{e}$. (Taking limits only preserves weak inequalities, which is why we get $\leq$ and not $<$.)

For the lower bound, compute a few terms:

$$\mathrm{P}(1) = \frac{3}{2}, \qquad \mathrm{P}(2) = \frac{3}{2} \cdot \frac{5}{4} = \frac{15}{8}, \qquad \mathrm{P}(3) = \frac{15}{8} \cdot \frac{9}{8} = \frac{135}{64} > \frac{128}{64} = 2.$$

Since the sequence is increasing, $L \geq \mathrm{P}(4) > \mathrm{P}(3) > 2$. Together,

$$2 < L \leq \mathrm{e}.$$



Checks

- $n = 1$: $(1+1)^1 = 2 < \mathrm{e}$ and $\mathrm{P}(1) = 1.5 < 2$. For $n = 4$: $\mathrm{P}(4) = \frac{2295}{1024} \approx 2.241 < \left(\frac{5}{4}\right)^4 \approx 2.441$.
- Computing the product numerically, $\mathrm{P}(n)$ settles at $L \approx 2.3842$ (by $n = 60$ it is constant to many decimal places), which is comfortably between 2 and $\mathrm{e} \approx 2.7183$.

Common slips

- Claiming $L < \mathrm{e}$ just because each $\mathrm{P}(n) < \mathrm{e}$. Strict inequalities can become equalities in the limit; only $L \leq \mathrm{e}$ follows. (Here $L < \mathrm{e}$ is in fact true, but it needs a separate argument.)
- Showing $\mathrm{P}(n) > 2$ for some n but forgetting to say that the sequence is increasing, which is what passes the bound on to L.
- Applying AM–GM without checking that the a_k are positive.

Beyond STEP

The step "increasing and bounded above, so convergent" is the monotone convergence theorem, which is equivalent to the completeness of the real numbers. Proving it from first principles is one of the first things a university analysis course does. Taking logarithms turns $\mathrm{P}(n)$ into the series $\sum \ln\left(1+2^{-k}\right)$, which converges by comparison with $\sum 2^{-k}$. That is the standard way to study infinite products. Abbott's *Understanding Analysis* treats both ideas beautifully.

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